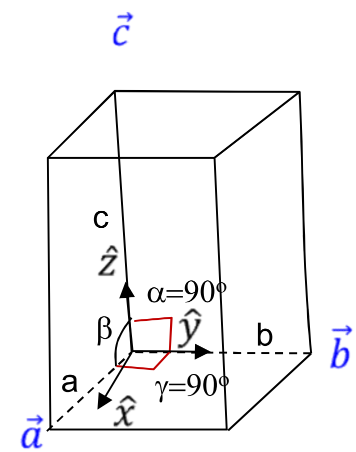
In-class exercise 3 (solution)

Consider a monoclinic lattice: $\Gamma^* = \begin{bmatrix} a^{*2} & a^*b^*\cos\gamma^* & a^*c^*\cos\beta^* \\ a^*b^*\cos\gamma^* & b^{*2} & b^*c^*\cos\alpha^* \\ a^*c^*\cos\beta^* & b^*c^*\cos\alpha^* & c^{*2} \end{bmatrix}$

Reciprocal lattice parameters {a*, b*, c*, 90°, β*, 90°}

Real space lattice parameters {a, b, c, 90°, β, 90°}



{a, b, c, α, β, γ}

$$\Gamma^*_{monocl.} = \begin{bmatrix} a^{*2} & 0 & a^*c^*\cos\beta^* \\ 0 & b^{*2} & 0 \\ a^*c^*\cos\beta^* & 0 & c^{*2} \end{bmatrix}$$

$$\vec{a}^* = \frac{\vec{b}\times\vec{c}}{V} \qquad V = \vec{a}\cdot(\vec{b}\times\vec{c}) = \vec{a}\cdot(\vec{y}\times\vec{z})bc\sin 90° = \vec{a}\cdot\vec{x}\,bc = abc\cos(90-\beta) = abc\sin\beta$$

$$\vec{a}^* = \frac{\vec{b}\times\vec{c}}{V} = \frac{bc\sin 90°}{abc\sin\beta}\hat{b}\times\hat{c} = \frac{1}{a\sin\beta}(\hat{b}\times\hat{c})$$

$$\vec{b}^* = \frac{\vec{c}\times\vec{a}}{V} = \frac{ca\sin\beta}{abc\sin\beta}\hat{c}\times\hat{a} = \frac{1}{b}(\hat{c}\times\hat{a})$$

$$\vec{c}^* = \frac{\vec{a}\times\vec{b}}{V} = \frac{ab\sin 90°}{abc\sin\beta}\hat{a}\times\hat{b} = \frac{1}{c\sin\beta}(\hat{a}\times\hat{b})$$

$$\Gamma^*_{monocl.} = \begin{bmatrix} \frac{1}{a^2\sin^2\beta} & 0 & \frac{-\cos\beta}{ac\sin^2\beta} \\ 0 & \frac{1}{b^2} & 0 \\ \frac{-\cos\beta}{ac\sin^2\beta} & 0 & \frac{1}{c^2\sin^2\beta} \end{bmatrix}$$

$$\cos\beta^* = \cos\pi-\beta$$
$$= \cos\pi\cos\beta + \sin\pi\sin\beta = -\cos\beta$$